Customer Transaction Analysis

Data Analytics Project Report

Dataset: 50,000 card transactions, 1 January to 14 October 2023

Tools: Python (Pandas, NumPy, Matplotlib, SciPy), SQL, Excel, Power BI

Prepared by: Gautam Kurpatwar

GitHub: github.com/GK3210

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1. Executive Summary

This project analyses 50,000 customer card transactions worth $22.11 million, recorded over 287 days in 2023, to understand where revenue comes from and how spending varies by category, customer group and time. The work covers data cleaning, feature engineering, exploratory analysis, statistical testing, SQL, charts and a dashboard.

Headline results

- Travel generates 58.4% of revenue and Electronics 19.9%, together 78% of the total.
- Transaction counts are nearly equal across the six categories (about 8,300 each), so revenue differences come from ticket size: Travel averages $1,540 against $55 for Restaurant.
- The top 10% of transactions produce 47.8% of revenue, and all 4,304 transactions above $1,500 are Travel purchases.
- Gender, age, month and weekday show no statistically significant effect on spend (all p-values above 0.2).
- Every Customer ID appears once, so repeat-purchase, retention and lifetime value analysis are not possible with this dataset.

Dashboard overview



Figure 1: Customer transaction dashboard (KPI cards and key charts)

2. Business Problem and Objectives

A retail or payments business needs to know which categories and customer segments drive revenue so that marketing, product and risk teams can prioritise effort. Transaction data is the most direct record of customer behaviour, but raw rows do not answer business questions until they are cleaned, summarised and tested.

Questions answered

1. How much revenue is generated, and which categories drive it?
2. Do spending patterns differ by gender or age group?
3. Is there a monthly or weekday pattern?
4. How concentrated is revenue in large transactions and merchants?
5. Are the differences we see statistically meaningful, or just noise?
6. How clean is the data, and what can it not tell us?

3. Dataset Description

The dataset has 50,000 rows and 9 columns, with no duplicate records.

Column	Description
Customer ID	Customer identifier (50,000 unique values)
Name, Surname	Customer name
Gender	F or M; 5,047 values missing
Birthdate	1948-11-02 to 2005-10-19 (customers aged 17 to 74 at transaction)
Transaction Amount	$5.01 to $2,999.88
Date	2023-01-01 to 2023-10-14 (287 distinct days)
Merchant Name	36,939 distinct merchants
Category	Restaurant, Market, Travel, Electronics, Clothing, Cosmetic

4. Methodology

The analysis followed a standard analytics workflow so that every number in this report can be reproduced from the code in the repository.

1. Understand: profile columns, types, ranges and missing values.
2. Clean: standardise column names, treat missing values, check for duplicates and invalid amounts.
3. Engineer features: age at transaction, age group, month, weekday, weekend flag and amount band.
4. Summarise: aggregate revenue, transaction count and average ticket by each dimension.
5. Visualise: build charts and a one-page dashboard.
6. Validate: test whether differences between groups are statistically significant.
7. Interpret: translate findings into recommendations and state the limitations.

5. Data Cleaning and Quality Assessment

Check	Result	Action
Missing values	Only Gender: 5,047 rows (10.1%)	Labelled Unknown, rows kept
Duplicate rows	None	No action
Zero or negative amounts	None (minimum $5.01)	No action
Date validity	All dates within 2023-01-01 to 2023-10-14	No action
Category labels	6 consistent labels, no spelling variants	No action
Customer age	17 to 74 years, plausible	No action
Text fields	Possible stray spaces	Trimmed

Gender was labelled Unknown rather than dropped so that revenue totals stay complete. The Unknown group has the same category mix and average ticket ($434) as the known groups, so keeping it does not bias results.

Features created

- Age at transaction (transaction date minus birthdate) and age group: 17-24, 25-34, 35-44, 45-54, 55-64, 65+.
- Month, weekday and a weekend flag from the transaction date.
- Amount band: Low (under $100), Medium ($100-500), High ($500-1,500), Premium (over $1,500).

The cleaned dataset is saved as data/processed/cleaned_transactions.csv and is ready to load into Power BI.

6. Key Metrics

KPI	Value
Total transactions	50,000
Total revenue	$22.11 million
Average transaction	$442.12
Median transaction	$182.20
Distinct merchants	36,939
Days covered	287
Average transactions per day	about 174
Average revenue per day	about $77,000
Highest revenue day	1 July 2023 ($115,721)

The mean is more than double the median, which shows that spend is strongly right-skewed by a small number of large purchases.

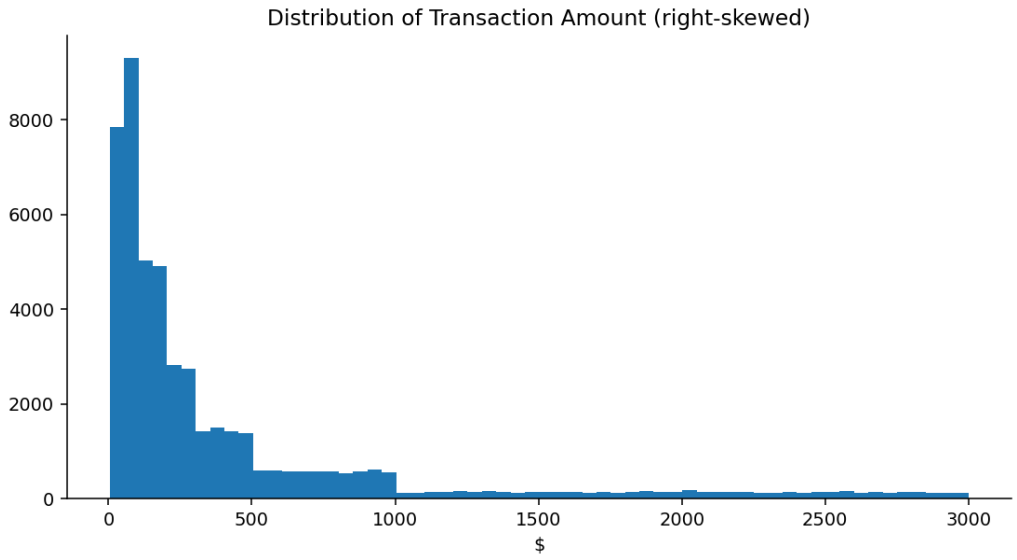


Figure 2: Distribution of transaction amounts

7. Analysis and Findings

7.1 Revenue by category

Category	Transactions	Revenue	Share	Avg ticket
Travel	8,377	$12.90M	58.4%	$1,540
Electronics	8,324	$4.39M	19.9%	$528
Market	8,382	$2.15M	9.7%	$257
Clothing	8,261	$1.32M	6.0%	$160
Cosmetic	8,243	$0.88M	4.0%	$106
Restaurant	8,413	$0.46M	2.1%	$55

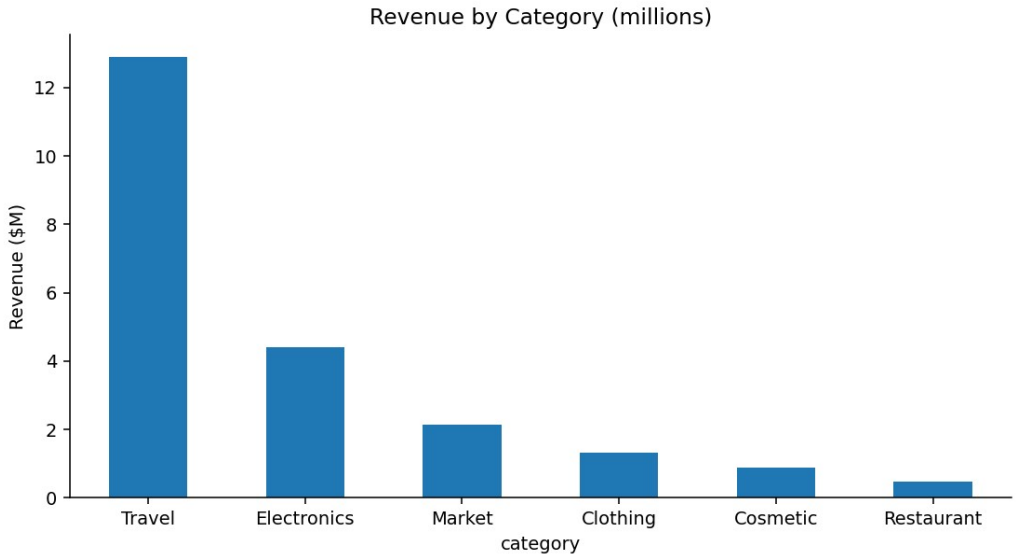


Figure 3: Revenue by category

Revenue is driven by ticket size, not volume. Travel and Electronics are the two categories that matter most commercially. Each category also sits in a distinct, bounded price range:

Category	Minimum	Q1	Median	Q3	Maximum
Restaurant	$10	$33	$55	$78	$100
Cosmetic	$10	$60	$107	$154	$200
Clothing	$20	$88	$160	$230	$300
Market	$5	$131	$259	$380	$500
Electronics	$50	$292	$530	$763	$1,000
Travel	$100	$823	$1,537	$2,242	$3,000

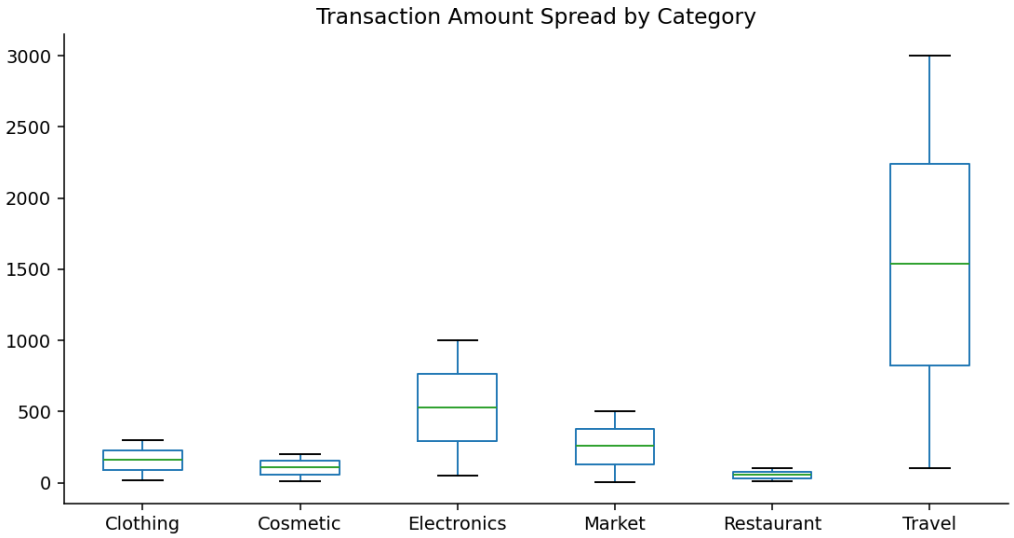


Figure 4: Spread of transaction amounts by category

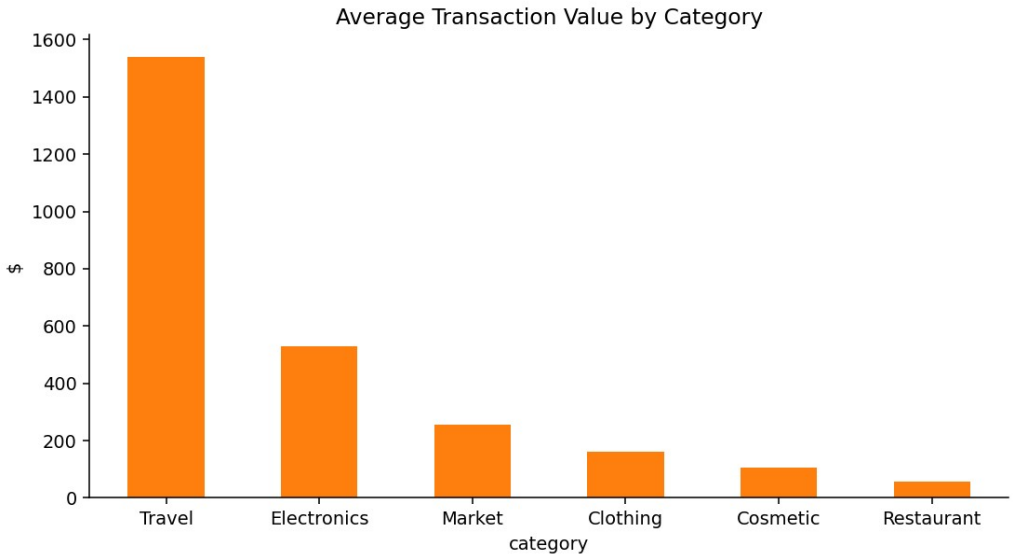


Figure 5: Average transaction value by category

7.2 Concentration in large transactions

Amount band	Transactions	Revenue	Revenue share
Low (under $100)	16,630	$0.94M	4.3%
Medium ($100-500)	21,732	$5.29M	24.0%
High ($500-1,500)	7,334	$6.24M	28.2%
Premium (over $1,500)	4,304	$9.63M	43.6%

The Premium band is 8.6% of transactions but 43.6% of revenue, and consists entirely of Travel purchases (51.4% of all Travel transactions exceed $1,500). Travel also makes up 62.1% of all transactions above $500. This is the segment with the highest value at risk from fraud, disputes or cancellation.

7.3 Gender

Group	Revenue share	Avg ticket	Transactions
Female	45.8%	$445.52	22,713
Male	44.3%	$440.42	22,240
Unknown	9.9%	$434.31	5,047

Average tickets differ by less than $12 across the three groups, and within each category the female and male averages are within about $20 of each other.

7.4 Age

Age group	Transactions	Revenue	Share	Avg ticket
17-24	7,045	$3.09M	14.0%	$438
25-34	8,571	$3.77M	17.1%	$440
35-44	8,519	$3.73M	16.9%	$438
45-54	8,517	$3.84M	17.4%	$451
55-64	8,639	$3.80M	17.2%	$440
65+	8,709	$3.87M	17.5%	$445

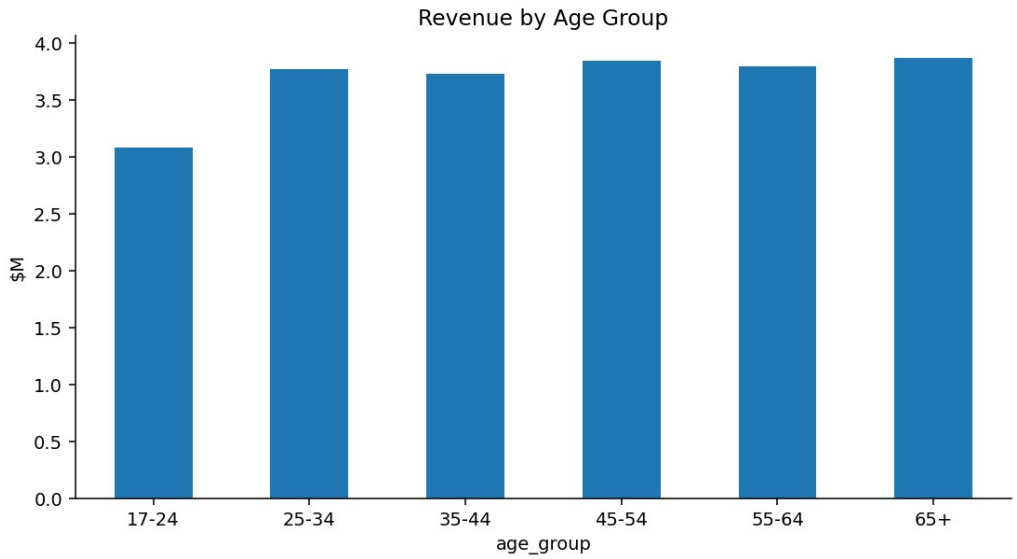


Figure 6: Revenue by age group

The 17-24 group has lower total revenue because it covers fewer birth years in the data, not because it spends less per transaction. The category mix is also similar across age groups.

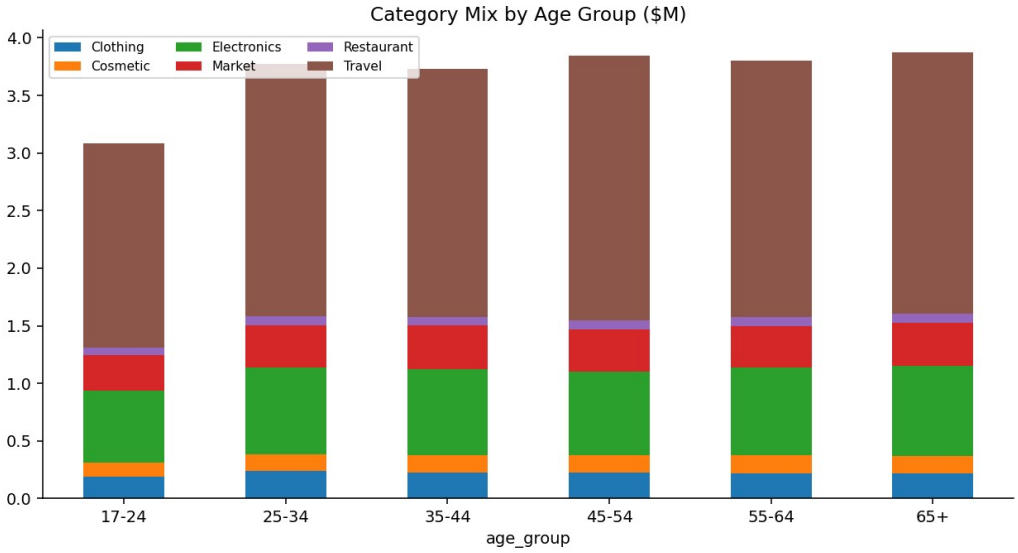


Figure 7: Category mix by age group

7.5 Time patterns

Month	Transactions	Revenue	Share	Avg ticket
January	5,444	$2.35M	10.6%	$431
February	4,854	$2.15M	9.7%	$443
March	5,456	$2.40M	10.8%	$439
April	5,091	$2.29M	10.4%	$450
May	5,493	$2.46M	11.1%	$448
June	5,258	$2.25M	10.2%	$429

Month	Transactions	Revenue	Share	Avg ticket
July	5,458	$2.48M	11.2%	$453
August	5,399	$2.39M	10.8%	$443
September	5,153	$2.27M	10.3%	$441
October (1-14 only)	2,394	$1.07M	4.8%	$447

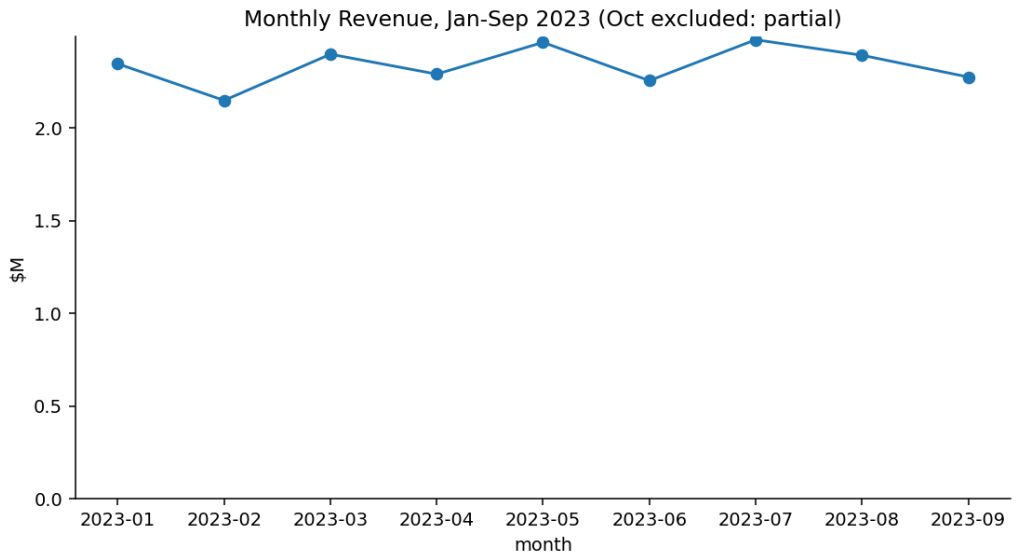


Figure 8: Monthly revenue, January to September 2023 (October excluded as a partial month)

Monthly revenue stays between $2.1M and $2.5M. February is lowest because it has fewer days. October covers only 1 to 14 October and should not be read as a decline.

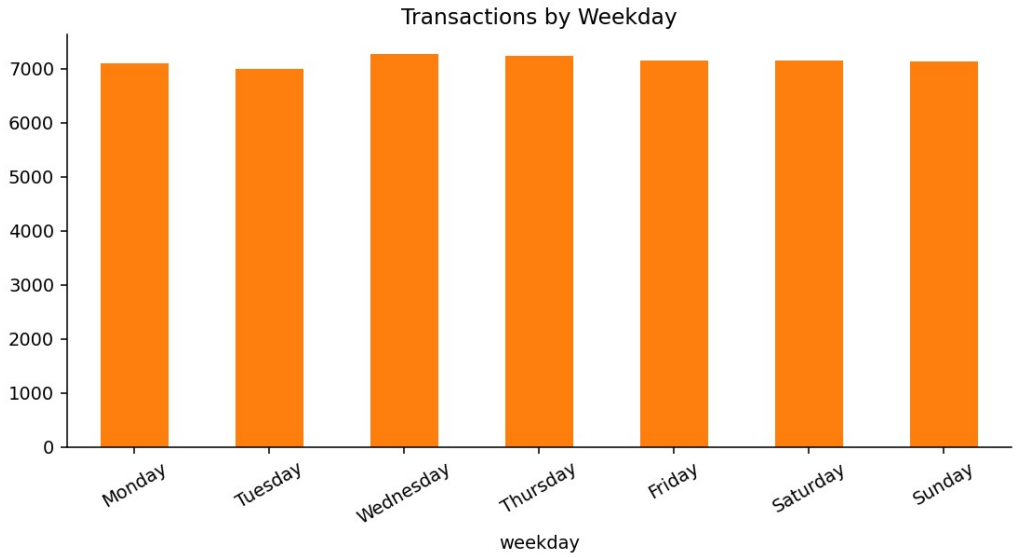


Figure 9: Transactions by weekday

Weekday volumes are flat. Monday has the highest average ticket ($459) and Tuesday the lowest ($432), but the statistical test in Section 8 shows this gap is within normal variation. Weekends account for 28.5% of transactions, close to the 2/7 expected from an even spread.

7.6 Merchants

There are 36,939 merchants for 50,000 transactions. A total of 32,907 merchants (89%) appear only once, only 33 have 30 or more transactions, and the ten largest merchants together account for just 1.3% of revenue. Merchant-level rankings are therefore not meaningful for decision-making in this dataset.

8. Statistical Validation

Visible differences between groups can be random noise, so each pattern was tested at the 5% significance level. The tests are in src/stats_tests.py.

Test	Method	p-value	Significant?
Spend by gender (F vs M)	Welch t-test	0.392	No
Spend by age group	One-way ANOVA	0.721	No
Spend by weekday	One-way ANOVA	0.219	No
Spend by month (Jan-Sep)	One-way ANOVA	0.515	No

The correlation between customer age and transaction amount is 0.003, which is effectively zero. None of the tested differences is statistically significant, so the visible gaps (for example Monday versus Tuesday) should not be used to drive decisions.

9. Business Insights and Recommendations

What the data says

- Revenue is determined by what is bought, not who buys it or when.
- A small share of large Travel and Electronics purchases carries most of the value.
- Demographic and calendar effects are statistically indistinguishable from noise.

Recommendations

1. Prioritise Travel and Electronics in promotions, partnerships and fraud monitoring, as they carry 78% of value.
2. Test cross-selling from Travel buyers into Electronics, and bundle low-ticket categories such as Restaurant and Cosmetic to raise basket size.
3. Avoid gender or age-targeted campaigns on this evidence, since spend is statistically identical across groups.
4. Add review rules for the Premium band (over $1,500), the highest value-at-risk segment.
5. Capture gender at sign-up to reduce the 10% missing rate.
6. Collect repeat-customer identifiers so retention and lifetime value can be measured.

10. Limitations

- Each Customer ID appears exactly once, so repeat purchase, retention, RFM and lifetime value analysis cannot be done.
- Amounts in each category fall within fixed ranges and the patterns are unusually uniform (equal category counts, flat spend by age and gender, randomly generated names). This

suggests synthetic or sample data, so the findings demonstrate method rather than real business behaviour.

- The data has no time of day, location, payment method or product detail.
- October is a partial month and is excluded from trend views.

11. Tools, Reproducibility and Project Structure

Python 3 with Pandas, NumPy, Matplotlib, SciPy and openpyxl was used for cleaning, analysis, testing and charts. SQL queries replicate the key questions, and the cleaned CSV is ready to load into Power BI. Running src/analysis.py, src/dashboard.py and src/stats_tests.py regenerates every table, chart and test in this report.

Path	Purpose
data/raw/	Original dataset (sample_dataset.csv)
data/processed/	Cleaned and enriched dataset
src/analysis.py	Cleaning, feature engineering, summary tables, charts
src/dashboard.py	Builds the one-page dashboard image
src/stats_tests.py	Statistical significance tests
sql/queries.sql	Seven SQL queries for the same questions
powerbi/POWERBI_GUIDE.md	DAX measures and dashboard layout
outputs/	Summary tables (Excel) and test results (CSV)
images/	Charts and dashboard PNG files
reports/	This report (DOCX and PDF)

12. Next Steps

1. Build the interactive Power BI dashboard using the supplied DAX measures and add slicers for category, gender, month and age group.
2. Repeat the analysis on data with repeat customers to add RFM segmentation, churn and lifetime value.
3. Add an anomaly detection step (for example z-scores by category) to flag unusual high-value transactions.
4. Automate the pipeline so that new monthly files are cleaned and reported without manual work.

13. Conclusion

Revenue in this dataset is determined by what is bought, not who buys it: Travel and Electronics dominate, a minority of large transactions carry nearly half of the value, and customer demographics and calendar effects add nothing that survives statistical testing. The most useful next step would be data that links purchases to repeat customers, which would enable segmentation, retention and lifetime value analysis.

Appendix A: SQL Queries

Selected queries from sql/queries.sql (MySQL style). The full file contains seven queries.

Revenue and share by category

```
SELECT category, COUNT(*) AS txns, ROUND(SUM(amount),2) AS revenue,
       ROUND(AVG(amount),2) AS avg_ticket,
       ROUND(100*SUM(amount)/SUM(SUM(amount)) OVER (),1) AS revenue_share_pct
FROM transactions GROUP BY category ORDER BY revenue DESC;
```

Monthly revenue with month-over-month change

```
WITH m AS (
  SELECT DATE_FORMAT(date,'%Y-%m') AS month, SUM(amount) AS revenue
  FROM transactions GROUP BY 1)
SELECT month, ROUND(revenue,0) AS revenue,
  ROUND(100*(revenue - LAG(revenue) OVER (ORDER BY month))
        / LAG(revenue) OVER (ORDER BY month),1) AS mom_pct
FROM m ORDER BY month;
```

Share of revenue from the top 10% of transactions

```
WITH r AS (SELECT amount, NTILE(10) OVER (ORDER BY amount DESC) AS decile
           FROM transactions)
SELECT ROUND(100*SUM(CASE WHEN decile=1 THEN amount END)/SUM(amount),1)
       AS top10_pct_share FROM r;
```